

33rd Conference On Neural Information Processing Systems (NeurIPS 2019)

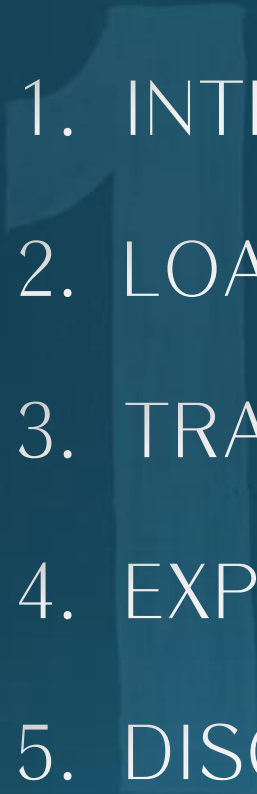
Towards Safe Online Reinforced Learning in Computer Systems

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1. INTRODUCTION
 2. LOAD BALANCING
 3. TRAINING WHEEL DESIGN
 4. EXPERIMENTS
 5. DISCUSSION

1.

Introduction

배경: 강화 학습 적용의 한계

- 상황 통제(control) 목적으로 다양한 영역에서 활용 증가
 - Congestion control (Jay et al., 2019)
 - Data processing cluster scheduling (Mao et al., 2018)
 - Routing (Valadarsky et al., 2017)
- 돌발 상황 발생 시 학습 정책(policy)의 강건함 확보는 아직 의문
- 실시간 학습 중 실수가 생기더라도 피해가 없게 만들면 된다 → 자전거 보조 바퀴 가설

강화 학습 모델의 자전거 보조 바퀴

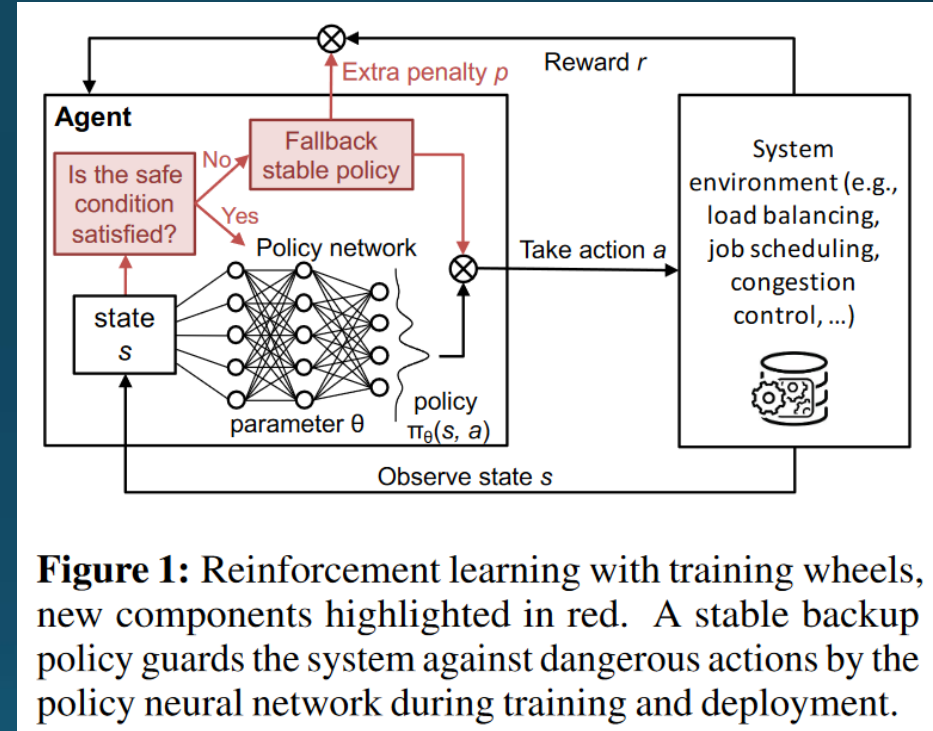
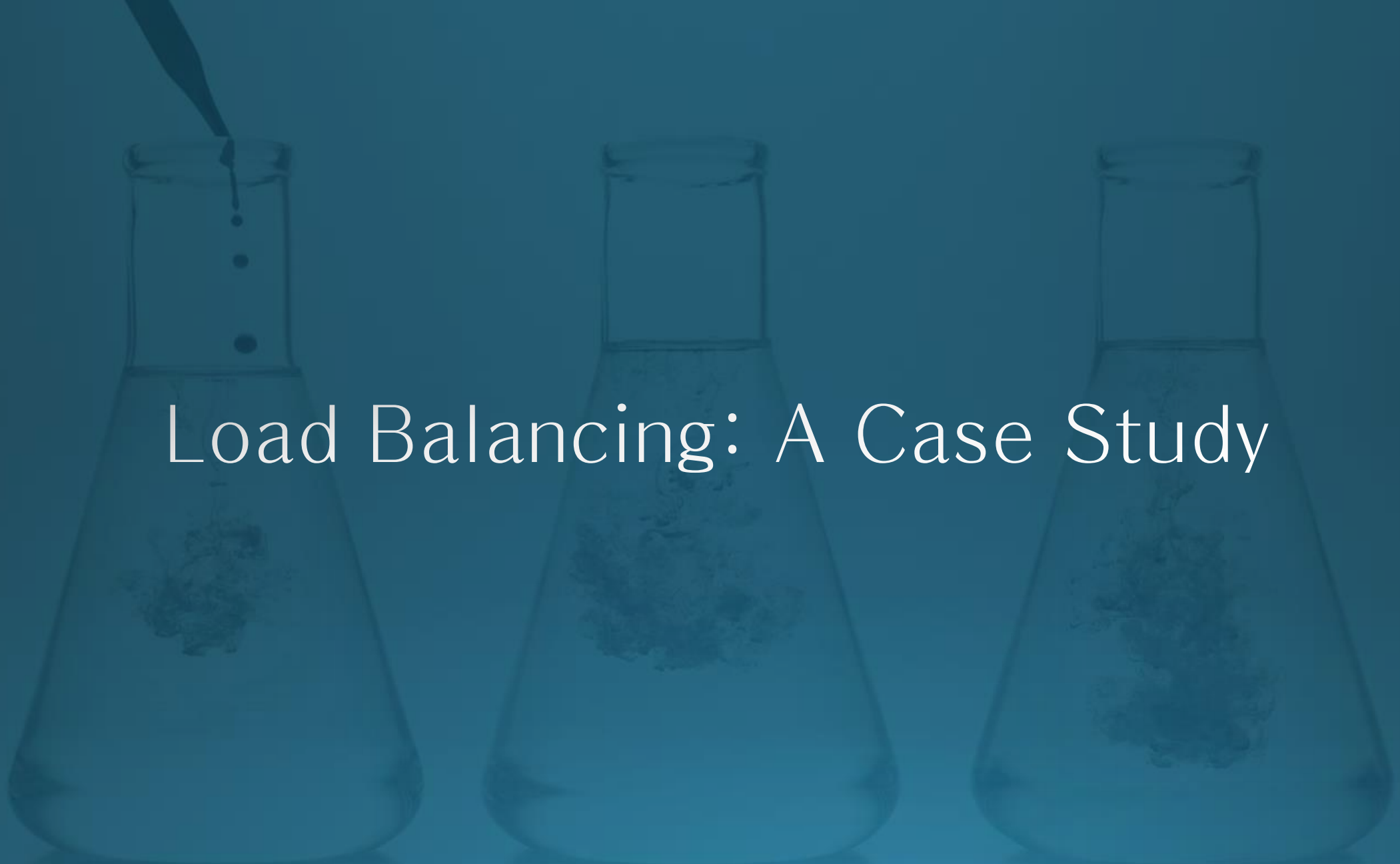


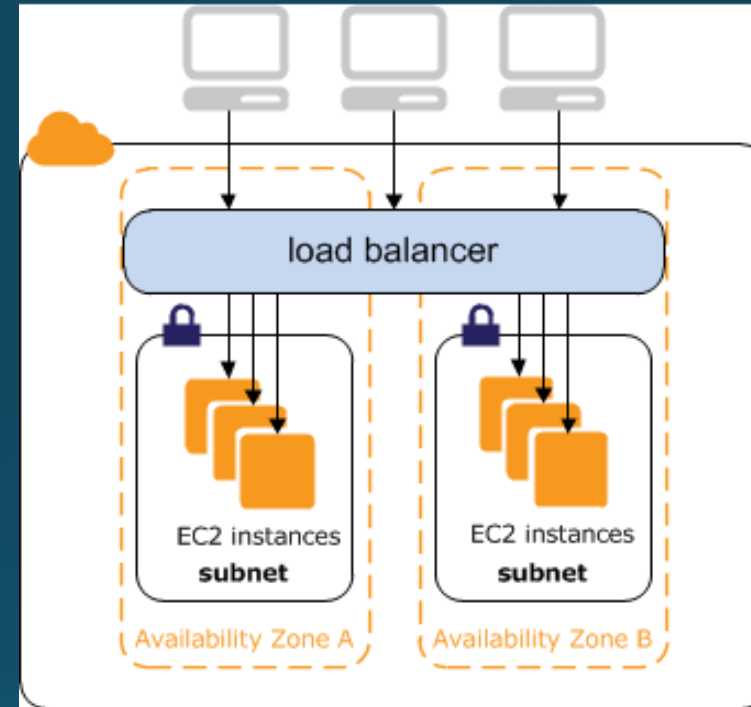
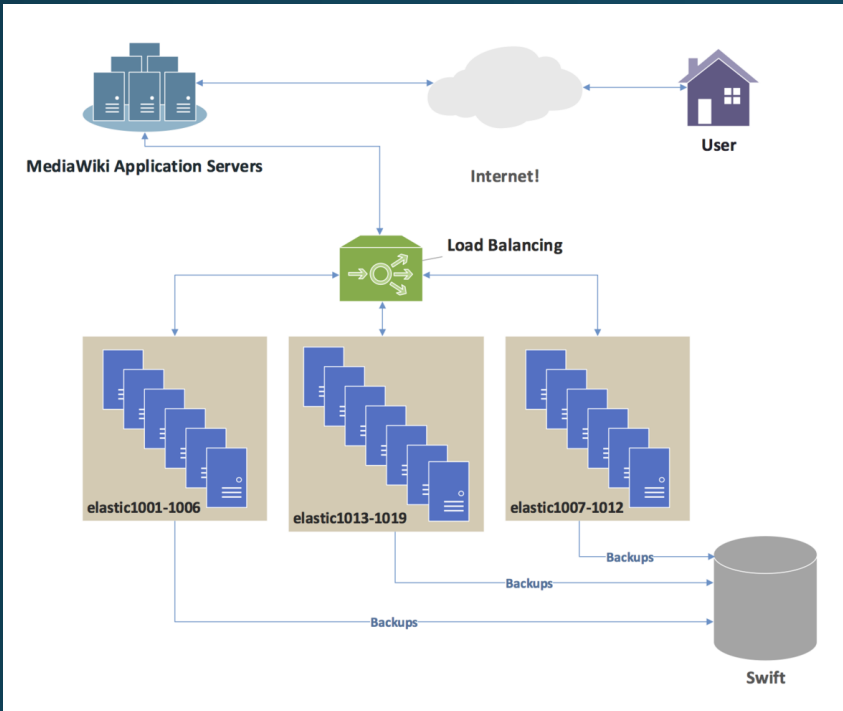
Figure 1: Reinforcement learning with training wheels, new components highlighted in red. A stable backup policy guards the system against dangerous actions by the policy neural network during training and deployment.

- 자전거를 타다 넘어지는 것을 막음 → fallback policy, guaranteed-safe policy
- 익숙해지면 자연스럽게 보조 바퀴를 쓰지 않고 주행 → 안정적인 학습 지속으로 성능 향상 도모

The background of the slide features three Erlenmeyer flasks arranged horizontally. A glass dropper is positioned above the leftmost flask, with a single drop of liquid falling into it. The flasks are partially filled with a dark, textured liquid. The entire scene is set against a solid blue background with a subtle gradient.

Load Balancing: A Case Study

강화 학습을 적용한 로드 밸런서 예시



- 서버에 가해지는 부하(workload) 분산(balance) 기능
- 실시간 부하 변동 따라 적절한 서버 선택 필요 → fixed policy 추종 RL 으로는 변동성 대응 ↓

RL Formulation (state, action, reward, penalty)

- **state** = {queue length at each server, the requests sent to server}
- **action** = $(a^J, a^L) \in \{1, 2, \dots, n\} \times \{l_1, l_2, \dots, l_k\}$

$n \rightarrow$ # of servers

$k \rightarrow$ # of discrete timeout choices

$l \rightarrow$ length of the queue on each server

$a^J \rightarrow$ request to server J

$a^L \rightarrow$ timeout for sending a hedged request to server L

- **reward** $r_i = -(t_i - t_{i-1}) \times M_i$

$i \rightarrow$ i^{th} step

$t_i \rightarrow$ wall time at step i

$M_i \rightarrow$ # of unfinished requests in the system

- **penalty** $= \sum_{i=1}^N (t_i - t_{i-1}) \times M_i$
- **Policy** $\pi = \text{minimize } (\sum_{i=1}^N (t_i - t_{i-1}) \times M_i)$

Q&A



The background of the slide features three Erlenmeyer flasks arranged horizontally. A pipette is positioned above the leftmost flask, with a few drops of liquid falling into it. The flasks contain a dark, textured substance, possibly a culture or sediment. The entire scene is set against a solid blue background.

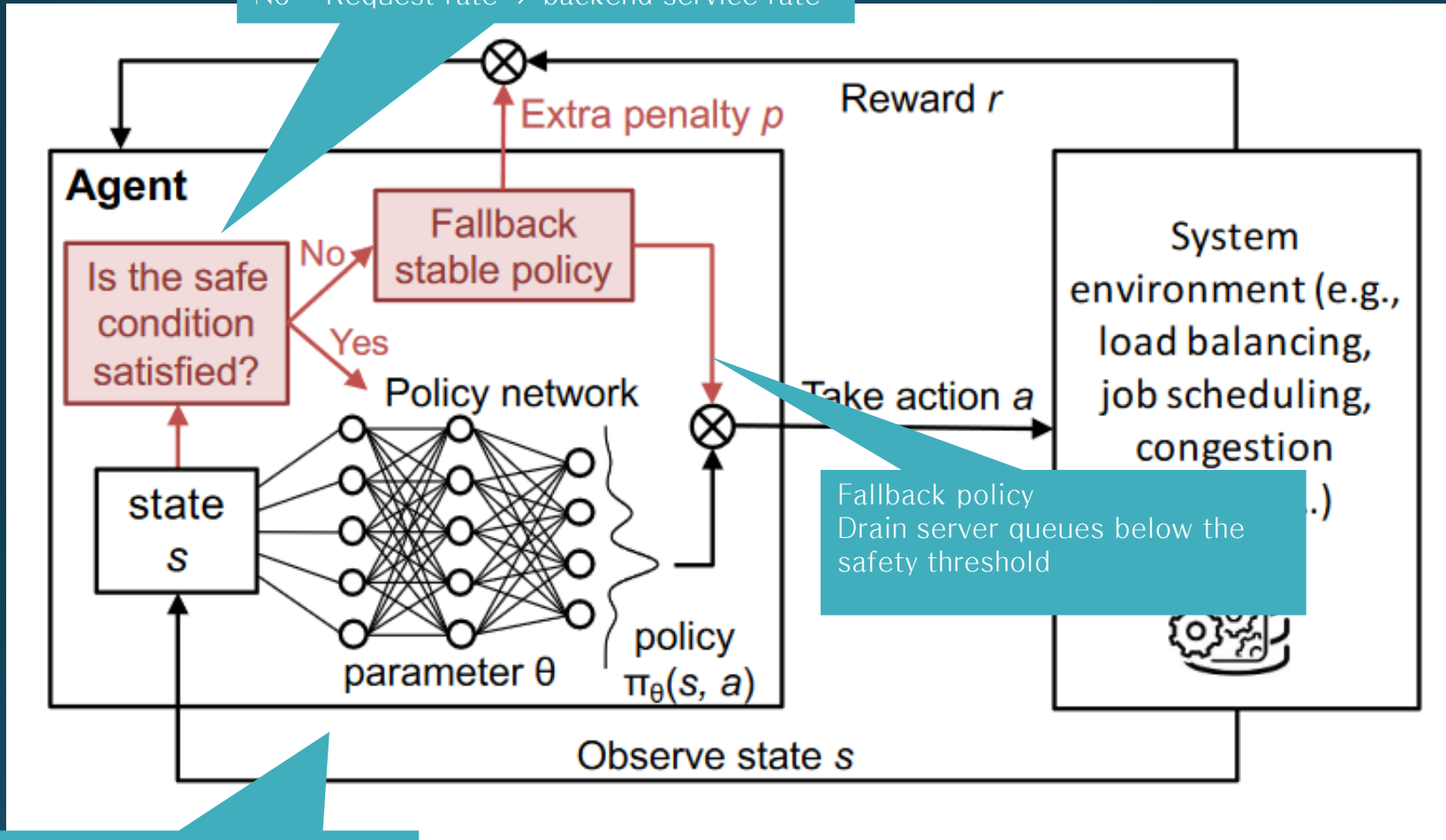
Training Wheels Design

로드 밸런서 사례

Safe condition $\rightarrow \max(Q_i) < 100$

Yes = Request rate < backend service rate

No = Request rate > backend service rate



Fallback policy
Drain server queues below the safety threshold

Network choose 1 of 7 timeout levels
2 hidden layers (# of neurons = 32, 16)
of parameters: 2,689
Batch size: 256

RL Formulation w/ fall back policy

- Updated policy π = minimize $(\underbrace{\sum_{i=1}^N (t_i - t_{i-1}) \times M_i}_{\text{Original objective}} + \underbrace{\beta \sum_{i=1}^N (t_i - t_{i-1}) \cdot 1}_{\text{Extra penalty}})$

$\beta \rightarrow$ extra penalty factor (e.g., 500)

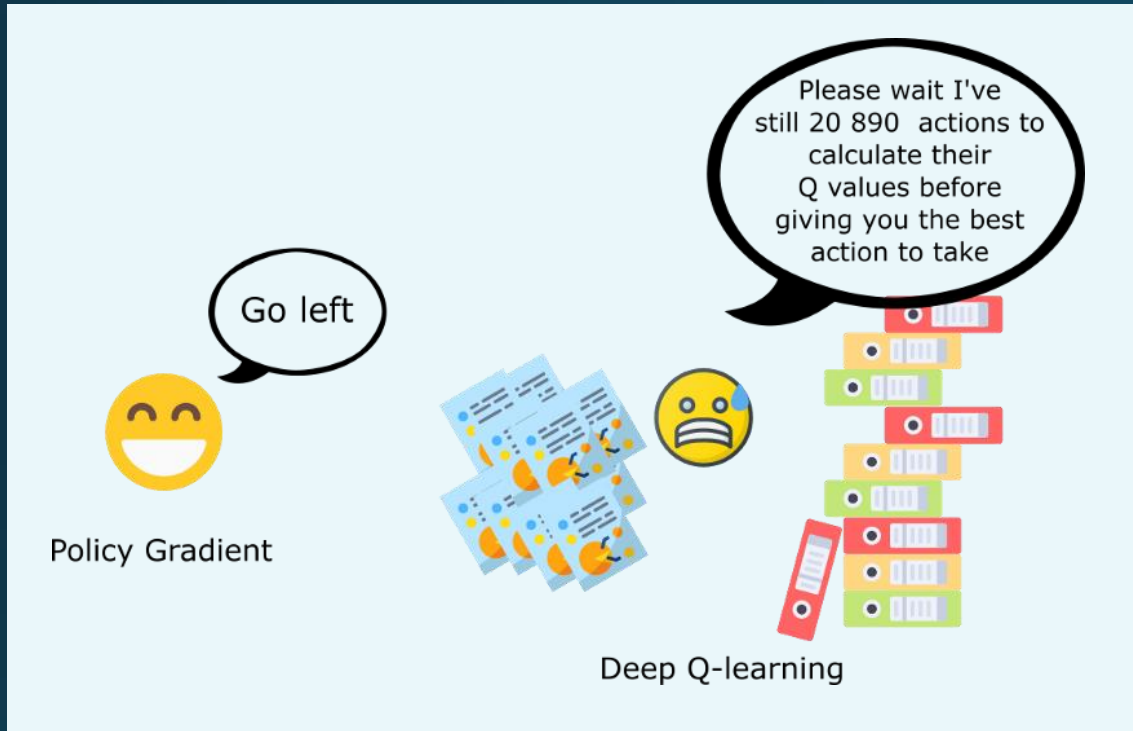
$i \rightarrow i^{\text{th}}$ step

$t_i \rightarrow$ wall time at step i

$M_i \rightarrow$ # of unfinished requests in the system

$1 \rightarrow$ indicates fallback policy (0 if in safe condition)

참고: Policy gradient method



- 수월한 강화학습 진척을 위해 정책(policy) 자체를 매개변수화(parameterize)
- 개별 action, state, reward 값 조절보다 원하는 학습 결과를 더 빨리 도출
- entropy factor 이용 gradient 기울기 조절
→ agent 다양한 탐색(exploration) 유도

저자들이 사용한 구체적인 내용은 아래 논문에...

- Policy Gradient Methods for Reinforcement Learning with Function Approximation, Sutton et al., NIPS 1999
- Asynchronous Methods for Deep Reinforcement Learning, Minh et al., ICML 2016
- High-dimensional continuous control using generalized advantage estimation, Schulman et al., ICLR 2016



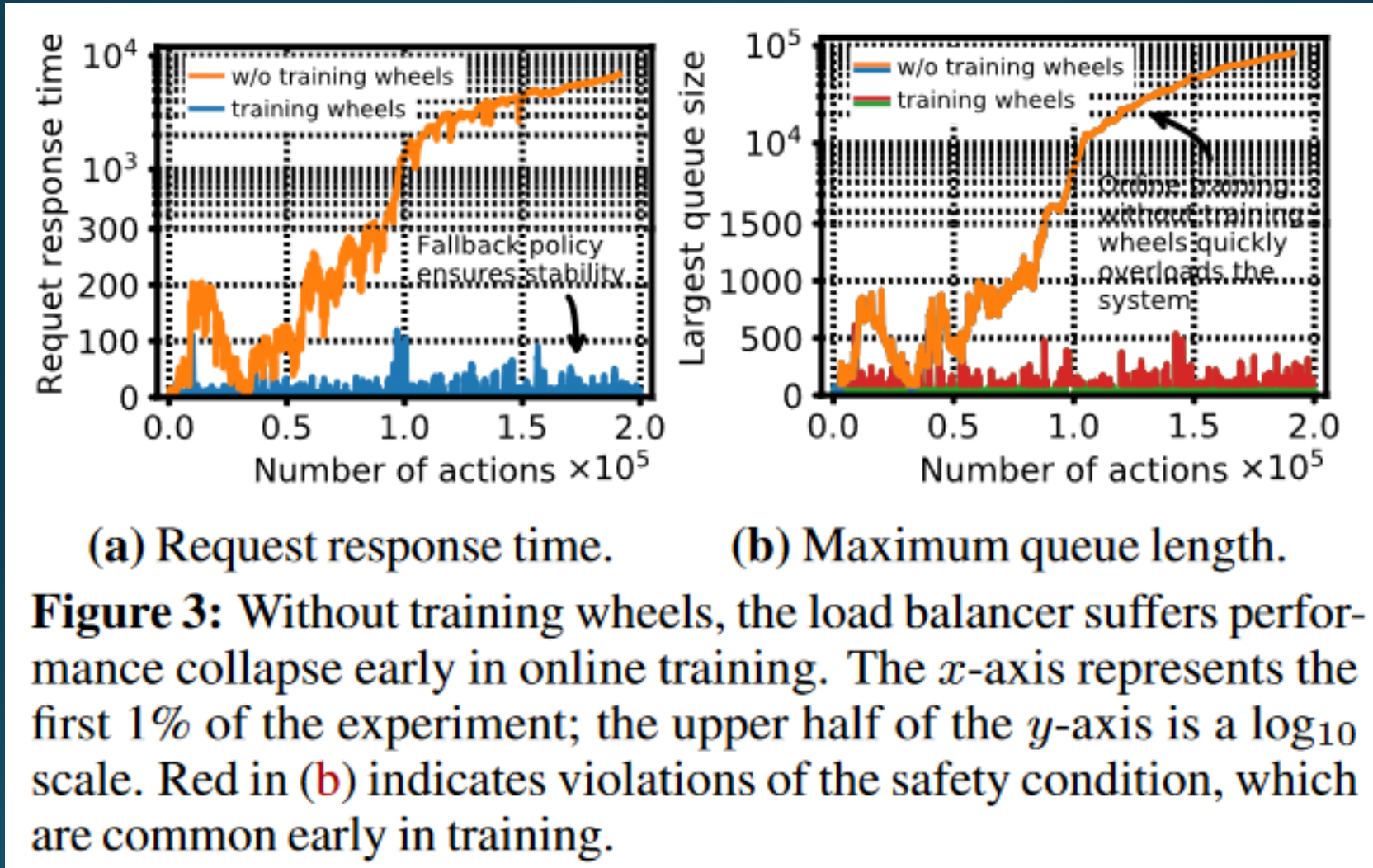
Experiments

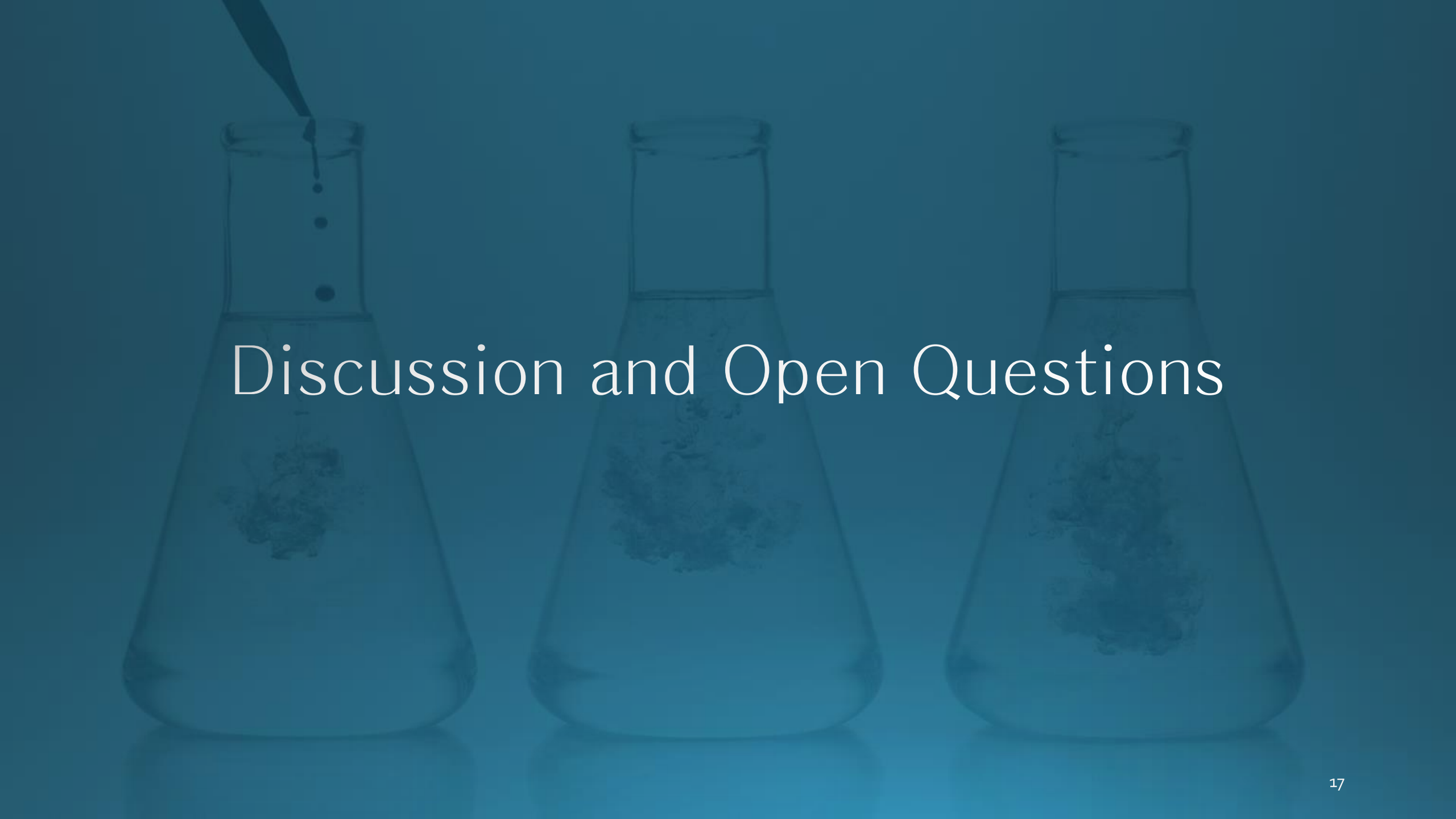
결과 1: 실시간 학습 시 부하 처리



Figure 2: A load balancer agent with training wheels is robust to workload changes, which occur at each vertical dashed line on the time series. When the largest queue exceeds 100 requests (horizontal dashed line), the fallback policy engages and drains the overloaded queue (red spikes). With this safety guarantee, request response time remains below the 100ms and the agent gradually converges to a new policy (entropy decreases) for all workloads.

결과 2: Fixed policy RL과 비교



The background of the slide is a solid blue color. In the center, there is a faint, semi-transparent image of three Erlenmeyer flasks arranged in a row. The flask on the left has a pipette tip positioned above its neck, with a few small droplets of liquid falling into it. Each flask contains a dark, textured substance at the bottom. The text 'Discussion and Open Questions' is overlaid in the center of the image in a white, serif font.

Discussion and Open Questions

여전히 개선해야 할 부분들도 존재함

- Expressing safety conditions and fallback policies: 실제로 적용이 가능한 분야인지?
- Automatic exploration: entropy factor 외 다른 방법은? 자동화?
- Learning from the fallback policy: 비정상 상황에서의 학습 결과 반영은?

Q&A

